Integrals – 1

Substitutions: rationalisation, Euler, the half-angle substitution

1 сентября 2026 года

Theoretical minimum.

Def. A substitution reduces an integral to a class we can already handle. Three classes must be recognised instantly: $R(x, x^{p_1/q_1}, \dots)$ is rationalised by $x = t^N$ with N the common denominator of the exponents; $R\bigl(x, \sqrt{ax^2+bx+c}\bigr)$ is handled by Euler's substitutions; $R(\sin x, \cos x)$ by the half-angle substitution $t = \tan\frac{x}{2}$.

Theorem 1 (Euler's substitutions). Let $y = \sqrt{ax^2+bx+c}$. If $a > 0$, put $y = \pm\sqrt{a}\,x + t$; if $c > 0$, put $y = xt \pm \sqrt{c}$; if the trinomial has real roots x_1, x_2, put $y = t(x - x_1)$. In each case both x and y become rational functions of t.

Theorem 2 (half-angle substitution). For $t = \tan\frac{x}{2}$ one has $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$ and $dx = \frac{2\,dt}{1+t^2}$. If the integrand is even in the pair $(\sin x, \cos x)$, the substitution $t = \tan x$ is cheaper; if it is odd in $\sin x$, use $t = \cos x$, and symmetrically.

Fact 1. Integration by parts is not a trick but the identity $\int u\,dv = uv - \int v\,du$: the only question is what to call dv, and the answer is almost always the most unpleasant factor that can still be integrated.

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1. Derive all three Euler substitutions and explain when each of them is the cheapest. Compute $\int \frac{dx}{x\sqrt{x^2+4x+3}}$ and $\int \sqrt{x^2+1}\,dx$, the second one in two ways (an Euler substitution and a hyperbolic one), and compare the answers.

2. Derive the half-angle formulas and compute $\int \frac{dx}{2+\cos x}$, $\int \frac{dx}{1+\sin x}$ (Integration Bee 2018) and $\int \frac{dx}{\sin x+\tan x}$. Explain why in the third integral $t = \tan\frac{x}{2}$ beats the substitution $t = \cos x$.

3. (Integration Bee) Rationalising substitutions:

   a) $\int \left(1 + x^{1/2} + x^{1/3}\right)\left(1 + x^{-1/2} + x^{-1/3}\right) dx$ (2016);

   b) $\displaystyle\int \frac{dx}{5 + 4\sqrt{x} + x}$ (2015);

   c) $\displaystyle\int \frac{x + \sqrt{x}}{1 + \sqrt{x}}\, dx$ (2025);

   d) $\displaystyle\int \frac{dx}{x + 3\sqrt{x}}$ (2019).

4. (Integration Bee, speed drill) Compute:

   a) $\int x^2\sqrt{x^3 + 2}\, dx$ (2017);

   b) $\int x^3\sqrt{x^2 + 1}\, dx$ (2016);

   c) $\int \frac{x\, dx}{x^4+x^2+1}$ (2016);

   d) $\int \frac{dx}{x(x^2+1)}$ (2017);

   e) $\int \frac{x^5}{2+x^{12}}\, dx$ (2018);

   f) $\int \frac{dx}{\sqrt{x\sqrt{x}-x^2}}$ (2018).

5. (Integration Bee) Self-similar integrands: $\displaystyle\int \sqrt{x + \sqrt{x + \sqrt{x + \cdots}}}\, dx$ (2014), $\displaystyle\int \sqrt{x\sqrt{x\sqrt{x\cdots}}}\, dx$ (2020) and $\displaystyle\int_0^1 \sqrt{x^2 + x + \sqrt{x^2 + x + \sqrt{\cdots}}}\, dx$ (2026). First say precisely what the infinite radical means, and only then integrate.